

Assignment 4: ODEs

1. Develop a general MATLAB function named `runge_kutta` to solve the following system of m initial value ODE problems using the classical 4th order Runge-Kutta method.

$$\frac{d\mathbf{y}}{dx} = \mathbf{f}(x, \mathbf{y})$$

subject to initial conditions

$$\mathbf{y}_0 = \mathbf{y}(x_0)$$

where x is the scalar independent variable, $\mathbf{y}$ is the vector of m dependent variables that needs to be solved, and $\mathbf{f}$ is a vector function.

Additional information (for informational purposes only):

Note that in scalar notation, the above vector equation can be rewritten as

$$\frac{dy_1}{dx} = f_1(x, y_1, y_2, \dots, y_m)$$

$$\frac{dy_2}{dx} = f_2(x, y_1, y_2, \dots, y_m)$$

$\vdots$

$$\frac{dy_m}{dx} = f_m(x, y_1, y_2, \dots, y_m)$$

subject to initial conditions

$$y_{1,0} = y_1(x_0)$$

$$y_{2,0} = y_2(x_0)$$

$\vdots$

$$y_{m,0} = y_m(x_0)$$

Please refer to lecture notes and in-class exercises for details of implementation.

Test your implementation for the following two problems.

(i) A mass balance for a chemical in a completely mixed reactor can be written as:

$$V \frac{dc}{dt} = F - Qc - kVc^2$$

where V = volume (10 m^3), c = concentration (g/m^3), F = feed rate ($200 \text{ g}/\text{min}$), Q = flow rate ($1 \text{ m}^3/\text{min}$), and k = a second order reaction rate ($0.1 \text{ m}^3/\text{g}/\text{min}$). If $c(0) = 0$, solve this ODE until the concentration reaches a stable level and plot it. What is the steady state concentration? Compare your solution with the analytical solution obtained using `dsolve`.

Note: `dsolve` will only work if you directly use the numeric values for F , Q , V , and k .

(ii) The reaction $A \rightarrow B$ takes place in two reactors in series. The reactors are well mixed but are not at steady state. The unsteady state mass balance for each stirred tank reactor is shown below, where CA_0 is the concentration of A at the inlet of the first reactor. CA_1 is the concentration of A at the outlet of the first reactor (and inlet of the second). CA_2 is the concentration of A at the outlet of the second reactor. CB_1 is the concentration of B at the outlet of the first reactor (and inlet of the second). CB_2 is the concentration of B in the second reactor. τ is the residence time for each reactor, and k is the rate constant for reaction A to produce B . Explain the physical meaning of each equation using mass balance principles.

$$\begin{aligned}\frac{dCA_1}{dt} &= \frac{1}{\tau}(CA_0 - CA_1) - kCA_1 \\ \frac{dCB_1}{dt} &= -\frac{1}{\tau}CB_1 + kCA_1 \\ \frac{dCA_2}{dt} &= \frac{1}{\tau}(CA_1 - CA_2) - kCA_2 \\ \frac{dCB_2}{dt} &= \frac{1}{\tau}(CB_1 - CB_2) + kCA_2\end{aligned}$$

Initial conditions

$$CA_1(0) = 0$$

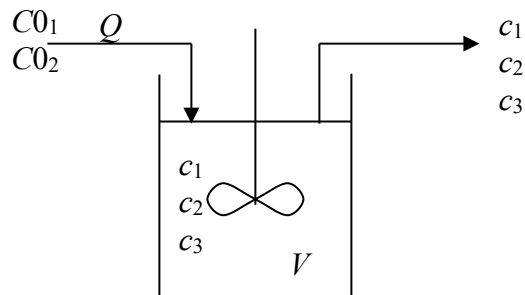
$$CA_2(0) = 0$$

$$CB_1(0) = 0$$

$$CB_2(0) = 0$$

If $CA_0 = 10$ mg/L, find and plot the concentrations of A and B in both reactors for the first 10 minutes of operation. Use $k = 0.1/\text{min}$ and $\tau = 5$ min. Use your own **runge-kutta** function and compare it with the results from the MATLAB's **ode45** function.

2. Two components, water (H_2O), and sulfur trioxide (SO_3) enter a completely stirred tank chemical reactor (CSTR). As they are mixed, they react to form a third product, sulfuric acid (H_2SO_4).



A simplified mathematical model for this process can be described by the following system of ODEs.

$$\begin{aligned}\frac{dc_1}{dt} &= \frac{Q}{V}(C0_1 - c_1) - \alpha_1 c_1 c_2 \\ \frac{dc_2}{dt} &= \frac{Q}{V}(C0_2 - c_2) - \alpha_2 c_1 c_2 \\ \frac{dc_3}{dt} &= -\frac{Q}{V}c_3 + \alpha_3 c_1 c_2\end{aligned}$$

where

c_1, c_2, c_3 = molar concentrations of components 1,2, and 3 in moles/m³.

Q = the volumetric flowrate in m³/min.

$\alpha_1, \alpha_2, \alpha_3$ = reaction rate constants for the decay or production of components 1,2, and 3.

$C0_1, C0_2$ = inflow concentrations of components 1 and 2 in moles/m³.

V = volume of the reactor in m³.

Solve the above system equations using the `runge_kutta` function you developed for $t = 0$ to 1.2 min (use 100 points in between). Verify the solution using MATLAB's `ode45`. Plot the solutions. Use the following conditions:

$$\text{At } t = 0, \quad c_1 = 5, \quad c_2 = 3, \quad c_3 = 0 \text{ moles/m}^3$$

$$C0_1 = 3.2, \quad C0_2 = 4.8 \text{ moles/m}^3$$

$$Q = 10 \text{ m}^3/\text{min}, \quad V = 2 \text{ m}^3$$

$$\alpha_1 = \alpha_2 = \alpha_3 = 2.6 \text{ m}^3/\text{mole-min}$$

What happens to the solution $\alpha_1 = 100000$, $\alpha_2 = 1$, and $\alpha_3 = 10$. Can you get reasonable solution with $n = 100$. If not, why?. How long does it take MATLAB's function `ode45` to obtain a solution?. Use the MATLAB function `ode23s` to obtain the solution.

Physical meaning of the above ODEs (additional information for understanding of the formulation – not needed to solve the problem):

The above equations can be derived using a simple mass balance for each component as follows:

rate of change of mass in the reactor for component i = mass inflow of component i – mass outflow of component i $\pm$ mass lost (–) or gained (+) due to the reaction in the tank.

Let's consider component 1. Rate of change of mass of component 1 = Vdc_1/dt . Mass inflow of component 1 = $QC0_1$. Since the tank is completely stirred, the outflow concentration is equal to the concentration in the tank. Therefore the mass outflow of component 1 = Qc_1 . If the rate at which the chemicals c_1 and c_2 react to form c_3 is proportional to the product of the two molar concentrations, then mass lost for component 1 due to reactions = $V\alpha_1 c_1 c_2$. This results in

$$V \frac{dc_1}{dt} = QC0_1 - Qc_1 - V\alpha_1 c_1 c_2$$

Similarly, the equations for the other components can be derived. For component 3, there is no mass influx and mass is produced (as opposed to lost) due the reactions.

3. A non-isothermal batch reactor can be described by the following stiff system of equations:

$$\frac{dC}{dt} = -e^{-10/(T+273)} C$$

$$\frac{dT}{dt} = 1000e^{-10/(T+273)} C - 10(T - 20)$$

Where C is the concentration of the reactant and T is the temperature of the reactor in °C. Initially, the reactor is at 15°C and has a concentration C of 1 gmol/L. Solve the above system using the following methods and compare the solutions for $t = 0$ to 10 s: (a) Your own Runge-Kutta, (b) MATLAB's `ode45` function. You may prepare two plots for this purpose: C vs t and T vs t .

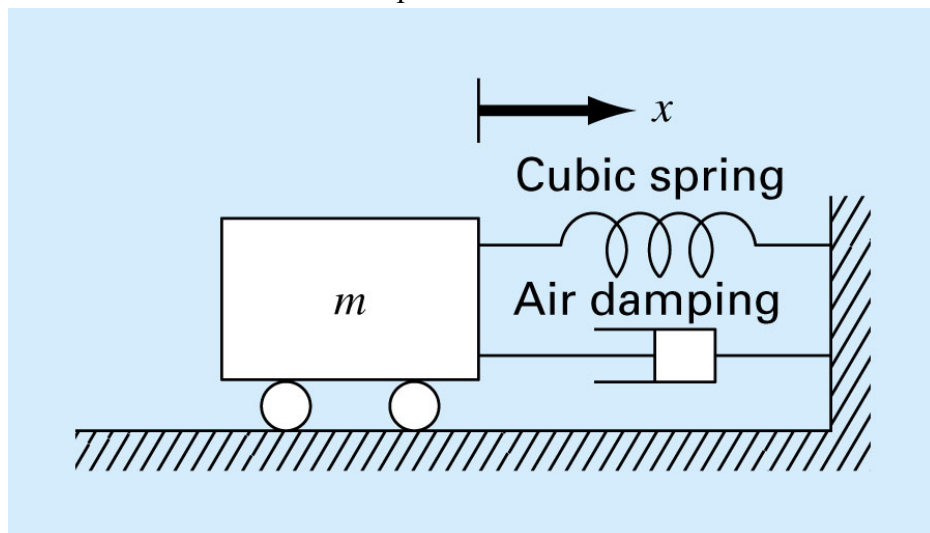
4. The following ordinary differential equation describes the motion of a damped spring-mass system shown.

$$m \frac{d^2 x}{dt^2} + (a) \text{sign} \left(\frac{dx}{dt} \right) \left(\frac{dx}{dt} \right)^2 + bx^3 = 0$$

where x = displacement from the equilibrium position, t = time, $m = 1$ kg mass, $a = 5$ N/(m/s)². The damping term is nonlinear and represents air damping. In this damping term, the function `sign()` determines the sign of the velocity since the velocity squared term is always positive. The spring is a cubic spring and is also nonlinear with $b = 5$ N/m². The initial conditions are:

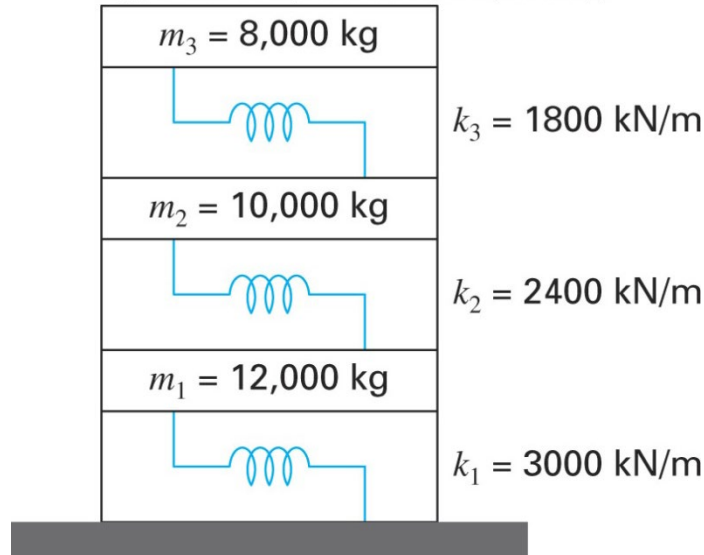
initial velocity: $\frac{dx}{dt} = 0.5$ m/s

initial displacement: $x = 1$ m



Reduce the above ODE to a system of two first order ODEs and then solve it using both your `runge-kutta` function and MATLAB's `ODE45` function for the time period $0 \leq t \leq 8$ s. Plot the displacement versus time and the velocity versus time for both cases.

5. Engineers and scientists use mass-spring models to gain insight into the dynamics of structures under the influence of disturbances such as earthquakes. The figure below shows such a representation for a three-storey building.



For this case, the analysis is limited to the horizontal motion of the structure. Using Newton's second law, force balances can be developed for this system as

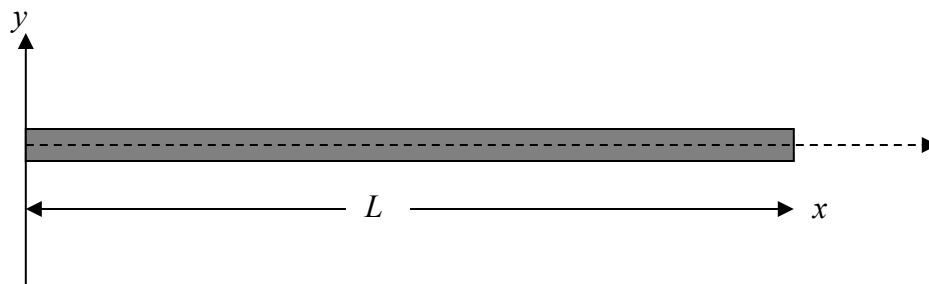
$$\begin{aligned}\frac{d^2 x_1}{dt^2} &= -\frac{k_1}{m_1} x_1 + \frac{k_2}{m_1} (x_2 - x_1) \\ \frac{d^2 x_2}{dt^2} &= \frac{k_2}{m_2} (x_1 - x_2) + \frac{k_3}{m_2} (x_3 - x_2) \\ \frac{d^2 x_3}{dt^2} &= \frac{k_3}{m_3} (x_2 - x_3)\end{aligned}$$

Simulate dynamics of the system from $t = 0$ to 20 s, given the initial condition that the velocity of the ground floor is $dx_1/dt = 1 \text{ m/s}$ and all other displacements and velocities are zero. Present your results as two time-series plots of (a) displacements and (b) velocities.

6. Deflection of a cantilever beam under its own weight can be described by the following ODE

$$EI \frac{d^2 y}{dx^2} = -\rho g \left(1 + \left(\frac{dy}{dx} \right)^2 \right)^{3/2} \left[x \left(x - \frac{L}{2} \right) + \frac{L^2}{2} \right]$$

where x is the distance along the beam, y is the deflection, L is the length of the beam, ρ



is the linear density, I is the moment of inertia, and E is the young's modulus. Solve the above equation using your `runge_kutta` function subject to the following boundary conditions: At $x = 0$, $y = dy/dx = 0$. Also use the following values: $L = 2$ m, $\rho g = 100$ N/m, and $EI = 2400$ Nm². Verify your solution using MATLAB's `ode45`. Plot both solutions. Find the deflection at $x = L$.

7. The deflection of a simply supported uniformly loaded beam (for small deflections) can be described by the following ODE.

$$EI \frac{d^2 y}{dx^2} = -\frac{q}{2}(x^2 - Lx)$$

subject to boundary conditions,

$$y(0) = 0, \quad y(L) = 0$$

where x is the distance along the beam, y is the deflection, L is the length of the beam, q is the load density, I is the moment of inertia, and E is the young's modulus. For a rectangular beam, $I = wh^3/12$ where w is the width and h is the height. Consider a wooden beam ($E = 20$ GN/m²), $L = 5$ m, 5 cm wide and 10 cm high, which is subjected to a uniform load of $q = 1500$ N/m on the 5 cm face.

(i) Solve this problem for $y(x)$ using centered finite differences. Use 100 points on the x -axis. You can first use finite differences to reduce the problem to a matrix equation $\mathbf{A}\mathbf{y} = \mathbf{b}$ and then solve using MATLAB's matrix solution operation $\mathbf{y} = \mathbf{A} \backslash \mathbf{b}$.

(ii) Verify your solution using the analytical solution obtained by integrating the above equation and then using the MATLAB function `bvp4c`.

When the load is applied on the 10 cm face, the deflections will be large. For large deflections, the ODE is non-linear and is described by

$$EI \frac{d^2 y}{dx^2} = -\frac{q}{2} \left(1 + \left(\frac{dy}{dx} \right)^2 \right)^{3/2} (x^2 - Lx)$$

(iii) Solve this large-deflection problem using MATLAB function `bvp4c` for the data given above. Note that $I = hw^3/12$ in this case.

(iv) Optional: Solve this problem again using central finite-differences and Newton's vector method and compare your solution with `bvp4c` solution in (iii).

HINTS:

The centered finite difference approximation for dy/dx and d^2y/dx^2 are given by

$$\frac{dy}{dx} \approx \frac{y_{i+1} - y_{i-1}}{2\Delta x}$$

$$\frac{d^2 y}{dx^2} \approx \frac{y_{i+1} - 2y_i + y_{i-1}}{\Delta x^2}$$